INCH-POUND

MIL-DTL-83505/1F
17 May 2004
SUPERSEDING
MIL-S-83505/1E
25 September 1991

DETAIL SPECIFICATION SHEET

SOCKETS (LEAD, ELECTRONIC COMPONENTS) (TYPE I, SOLDERLESS WRAP)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83505.

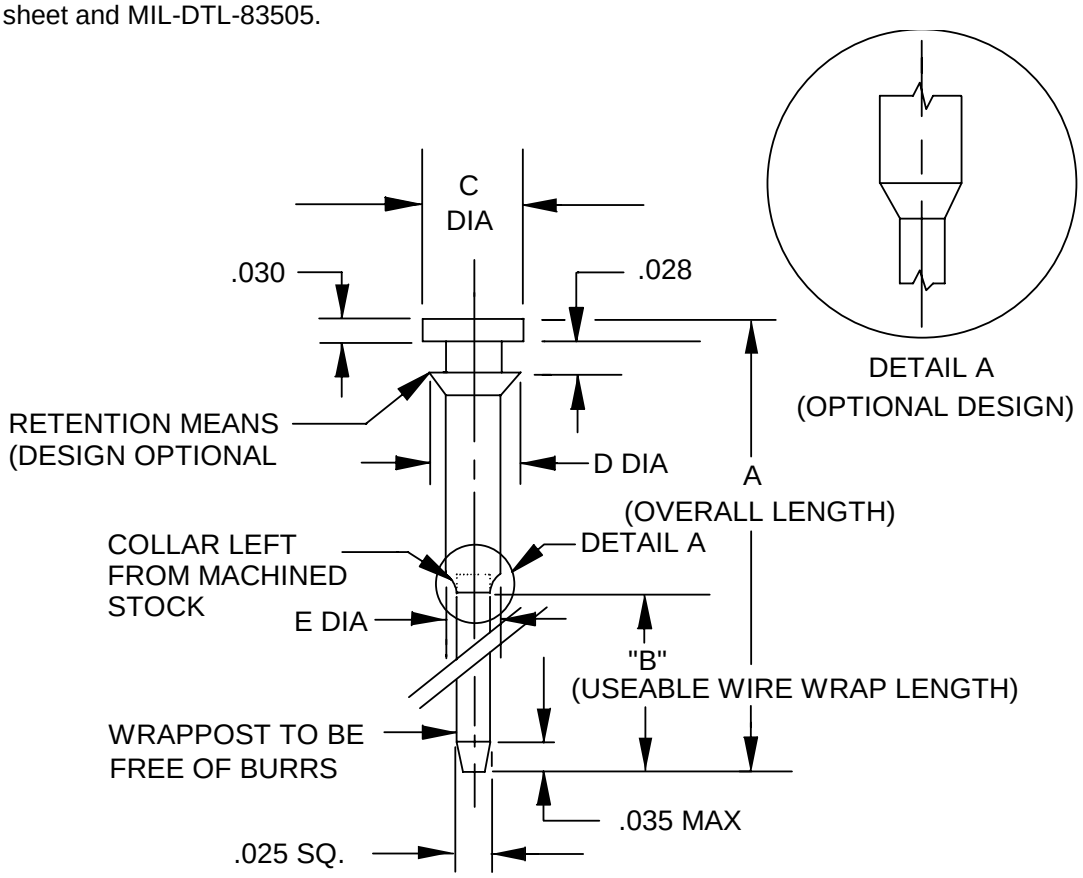


FIGURE 1. Socket, lead, (type I) solderless wrap.

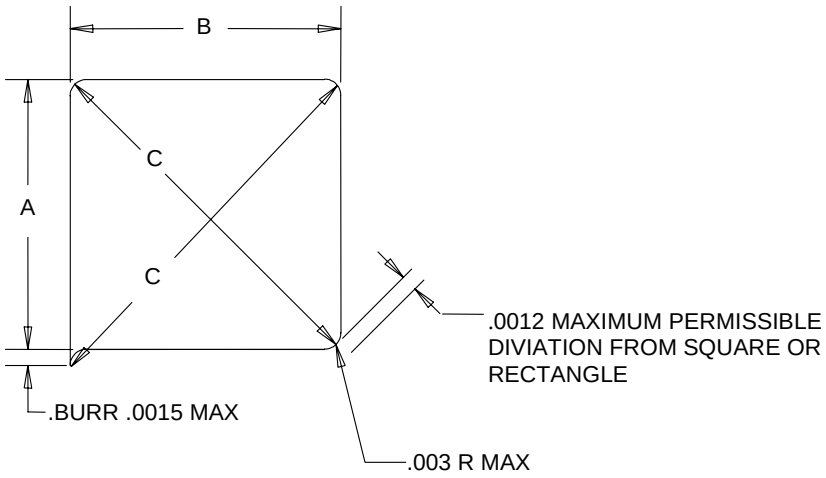
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Dash number	A ".010 (0.25)	B ".010 (0.25)	C Dia	D Dia ".002 (0.05)	E Dia
-001 -002	.703 (17.86)	.475 (12.07)	.072 (1.83)	.060 (1.52)	.052 (1.32)
-003 -004	.563 (14.30)	.335 (8.51)	.072 (1.83)	.060 (1.52)	.052 (1.32)
-005 -006	.453 (11.51)	.225 (5.71)	.072 (1.83)	.060 (1.52)	.052 (1.32)
-007 -008	.812 (20.62)	.475 (12.07)	.085 (2.16)	.082 (2.08)	.073 (1.85)
-009 -010	.703 (17.86)	.475 (12.07)	.063 ±.003 (1.60 "0.08)	.060 (1.52)	.052 (1.32)
-011 -012	.563 (14.30)	.335 (8.51)	.063 ±.003 (1.60 "0.08)	.060 (1.52)	.052 (1.32)
-013 -014	.453 (11.51)	.225 (5.71)	.063 ±.003 (1.60 "0.08)	.060 (1.52)	.052 (1.32)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified the tolerance for three place decimals is ± .005 inch (0.13 mm).
4. Part is to be free of burrs, except as noted.
5. Dimensions apply after plating.

FIGURE 1. Socket, lead, (type I) solderless wrap - Continued.

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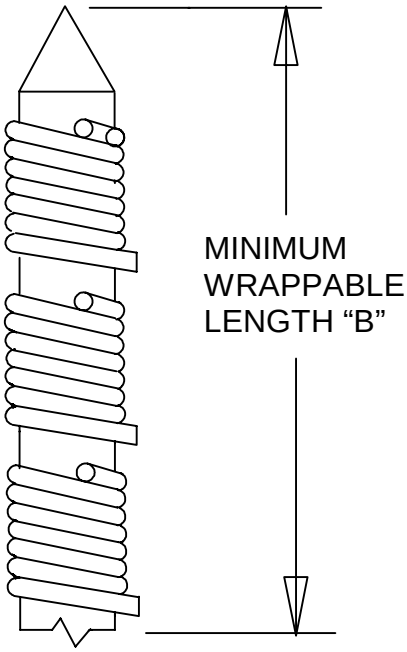
Inches	mm
.0012	0.030
.0015	0.038
.003	0.076

A (mm)	B (mm)	C (mm)	Parallelism (mm)	Straightness Inch/inch (mm)
.025 (0.63) nominal .022 (0.56) min	.025 (0.63) nominal .022 (0.56) min	.0355 (0.902) max .0325 (0.825) min	.002 (0.05)	.005 (0.13)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Wrappost geometry.

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MINIMUM WRAPPABLE LENGTH, -B, INCHES

(see note 5)

Number of wrapped connections	Wire gauge (AWG)						
	30	28	26	24	22	20	18
1	.185 (4.70)	.219 (5.56)	.226 (5.74)	.258 (6.55)	.303 (7.70)	.327 (8.31)	.391 (9.93)
2	.320 (8.13)	.388 (9.86)	.402 (10.21)	.466 (11.84)			
3	.455 (11.56)						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified the tolerance is ± .005 inch (0.13 mm).
5. Minimum wrappable length shall have the wrappost geometry in accordance with figure 2.

FIGURE 3. Wrappost configurations and wire gauge wrappable length.

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REQUIREMENTS

Qualification:

Group qualification: Shall be as specified in table I.

Dimensions and configuration: Shall be in accordance with figures 1, 2, and table II.

Solderless wrappost geometry: The solderless wrappost geometry shall be in accordance with figures 2 and 3.

TABLE I. Group qualification.

Dash number	Qualification tests
-001, -002, -009, and -010 (as applicable)	Require all qualification tests of MIL-DTL-83505.
-003 through –006 and -011 through -014 (as applicable)	Visual and mechanical tests of MIL-DTL-83505 and qualification of one of the following: -001, -002, -009, -010
-007 and -008 (as applicable)	Groups 1, 2, and 5 of qualification table of MIL-DTL-83505 and qualification of one of the following: -001, -002, -009, or -010.

TABLE II. Interface dimensions and finish.

Dash number	Recommended mounting hole dia (mm) ".001 (0.03)	Accepts pin		Sleeve plating
		Length (mm)	Diameter (mm)	
-001 and -009	.055 (1.40)	.175 (4.44)	.021 (0.53)	Gold
-002 and -010		.125 (3.18)	.016 (0.41)	Tin/lead
-003 and -011	.055 (1.40)	.175 (4.44)	.021 (0.53)	Gold
-004 and -012		.125 (3.18)	.016 (0.41)	Tin/lead
-005 and -013	.055 (1.40)	.175 (4.44)	.021 (0.53)	Gold
-006 and -014		.125 (3.18)	.016 (0.41)	Tin/lead
-007	.076 (1.93)	.281 (7.14)	.030 (0.76)	Gold
-008		.156 (3.96)	.020 (0.51)	Tin/lead

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Engagement (mating) and separation (unmating) forces shall be in accordance with MIL-DTL-83505 except as specified in table III.

TABLE III. Engagement (mating) force.

Dash numbers	Test gauge B (mm)	Mating force max lbs (kg)
-001 and -009	.125 (3.18)	1.25 (0.57)
-002 and -010		
-003 and -011	.125 (3.18)	1.25 (0.57)
-004 and -012		
-005 and -013	.125 (3.18)	1.25 (0.57)
-006 and -014		
-007 and -008	.156 (3.96)	4.00 (1.81)

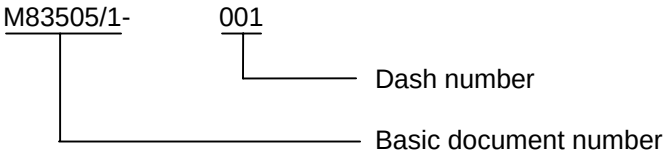
Terminals: Type I (solderless wrap).

Mounting: Use .046 inch (1.16 mm) minimum thickness mounting board.

Test mounting:

Use glass epoxy board, type G10 or FR4 material for mounting sockets.

Part or Identifying Number (PIN): The PIN shall consist of the basic number of this specification sheet and the dash number from table II.



Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Navy - EC
Army - CR
Air Force - 11
DLA - CC

Review activities:
Air Force - 99

Preparing activity:
DLA - CC

(Project 5935-4510-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil/.

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